Recommendation Systems

Recommendation systems are used all over the world to enhance the experience for users in streaming movies, listening to music, buying products, and many other applications. These systems use artificial intelligence (AI) to suggest something new to the user based on their past patterns and preferences (Murel and Kavlakoglu) .AI will analyze the past data of the user to make predictions about what they will like. The system will analyze everything from what the user interacts with, how long they interact with it, and the frequency in which they interact with the item. Based on this information, the system will make predictions about what the user would enjoy in the future. The goal in adding this was to make the viewing experience simpler for users and to increase the overall engagement of their application.

There are two main types of recommendation systems that exist. The first is content-based filtering which analyzes your personal history with the application to make recommendations. The second is called collaborative filtering which analyzes the similarities between users to make recommendations (Murel and Kavlakoglu). Both systems are used in many applications, and they are often used together. These systems have impacted the overall use of these applications, shaping how users interact with this content.

Content-based filtering is a system that is specifically based around the user. This system looks at everything the users does. This includes what the user looks at, how long the user interacts with the content, the specific things associated with that content, and many more key factors. When putting all of these features together, the system is able to build a profile to categorize what it thinks you like based on your past interactions (Abdurrafi and Ningsih).

Systematically, the algorithm profiles a user by putting the things they enjoy into different categories. These categories are then turned into numbers that are being compared to other numbers in the system. The cosine similarity algorithm is commonly used to identify like features(Permana, Permana, and Wibowo). This type of filtering can be useful because it can provide personalized content tailored to the user. However, with this type of system it can lack variation of suggestions for new content. Despite this setback, this system still manages to make the user feel like they are getting individualized results.

Collaborative filtering is different from content-based filtering because it is focused on all users. This system focuses on the behaviors of the users rather than focusing on the features of the content. In focusing on the user's behavior, it analyzes what content they and similar users interact with. Based on the behavior of users like you, it will recommend a similar user with new content they haven't interacted with yet. This system commonly seemed in places like Netflix or amazon where it says, "People who watched this also watched this" (Murel and Kavlakoglu)..

Cosine similarity is used in this type of filtering by assigning a number to each users preferences and behaviors. If you think of the numbers as vectors, you can measure how similar the behaviors are by measuring the angle between the vectors. Some of the benefits of using this method over content-based filtering is that it allows users to explore new content that they may not have been aware of (Singh et al.). One big limitation this has is that it isn't very effective for new users. Since these user's haven't interacted with any content, the system doesn't know what to suggest because it doesn't know what the person likes.

As mentioned above, the cosine similarity algorithm is used a lot in allowing recommendation systems to function the way they do. The main way this algorithm functions is by having a person's preferences or features represented by specific vectors in numeric form. For the system to correctly recommend content the user will like, the algorithm measures the angle between the vectors. The smaller the angle is, the more similar the items are. This will increase the likelihood of the system recommending content the user will enjoy (Singh et al.).

If we look at how recommendation systems are used in the real world, there are many practical applications for them in our everyday lives. Many companies like Netflix, Amazon, or Spotify all use these recommendation systems. Netflix will suggest shows and movies to viewers based on the type of entertainment the user is consuming. Amazon will suggest products to customers based on items they have bought before or products other users have bought. You can commonly see this when a customer shows interest in one piece of merchandise, it will suggest you buy another product because others have commonly bought these products together. In Spotify, this is also evident because it will recommend songs and playlists based on your listening habits. There is even a "DJ" function on Spotify where it will play songs based on information it has from your listening history (Murel and Kavlakoglu).

Our lesson is designed to give you more information on how recommendation systems work by showing the difference between collaborative and content-based filtering in an interactive way. During out presentation we will demonstrate how these methods analyze data to provide recommendations in real-life scenarios. We will also include a detailed explanation of how exactly the cosine similarity algorithm works. We also plan to include a detailed coding exercise to demonstrate how to algorithm works in a personalized way.

Works Cited

Juni Permana, Armadhani Hiro Juni Permana, and Agung Toto Wibowo. “Movie Recommendation System Based on Synopsis Using Content-Based Filtering With TF-IDF and Cosine Similarity”. *International Journal on Information and Communication Technology (IJoICT)*, vol. 9, no. 2, Dec. 2023, pp. 1-14.

Muhammad Falah Abdurrafi, and Dewi Handayani Untari Ningsih. “Content-Based Filtering Using Cosine Similarity Algorithm for Alternative Selection on Training Programs.” *Journal of Soft Computing Exploration*, vol. 4, no. 4, Dec. 2023, pp. 204–12.

Ph.D, Jacob Murel, and Eda Kavlakoglu. “Content-Based Filtering.” *Ibm.com*, 21 Mar. 2024.

Singh, Ramni Harbir, et al. "Movie recommendation system using cosine similarity and KNN." International Journal of Engineering and Advanced Technology 9.5 (2020): 556-559.